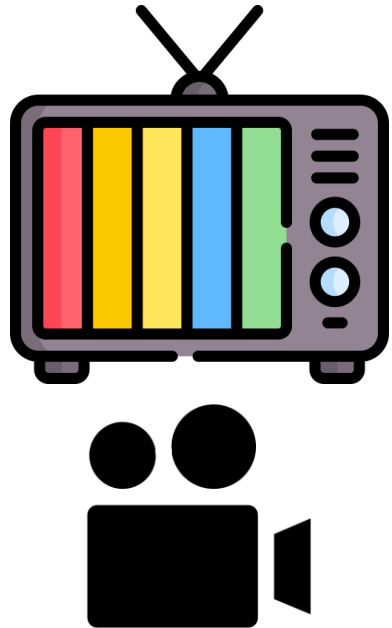


Content Analysis

Dr. Sofia Gil-Clavel



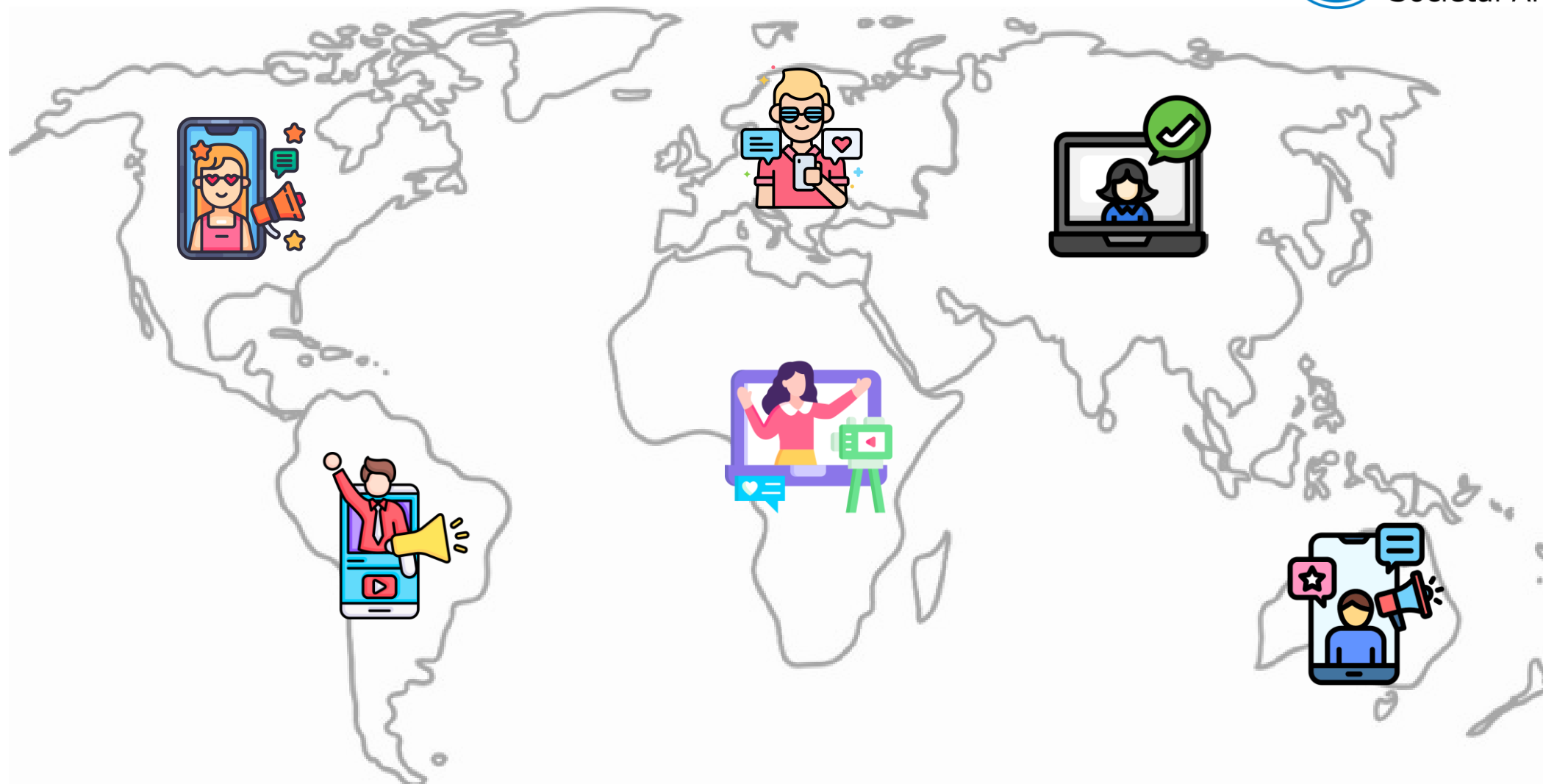


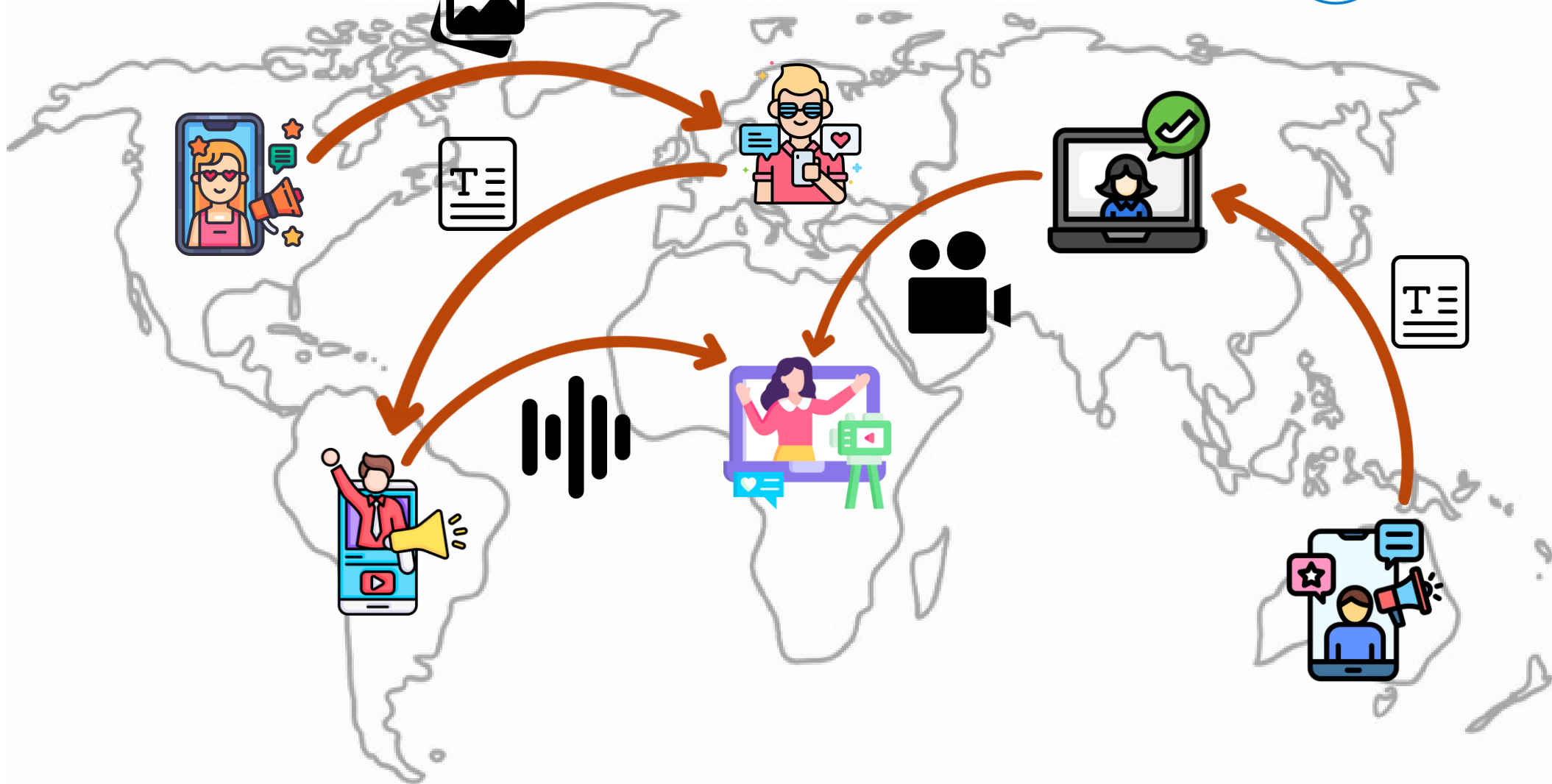
















How can we analyze this data?
Can we make sense of all this?

Content Analysis

How can we make use of all this data?



Content Analysis



“[...] analysis of the manifest and latent content of a body of communicated material (as a book or film) through:

classification,
tabulation, and
evaluation

of its key symbols and themes in order to ascertain its meaning and probable effect.”

Source: Krippendorff, Klaus. *Content Analysis: An Introduction to Its Methodology*. SAGE Publications, Inc., 2019. <https://doi.org/10.4135/9781071878781>.

Content Analysis

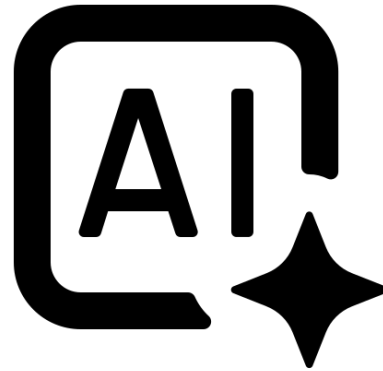


“[...] content analysts examine:
data,
printed matter,
images, or
sounds

in order to understand what they mean to people, what they enable or prevent, and what the information conveyed by them does.”

Source: Krippendorff, Klaus. *Content Analysis: An Introduction to Its Methodology*. SAGE Publications, Inc., 2019. <https://doi.org/10.4135/9781071878781>.

Imagine you want to analyze news about Artificial Intelligence.



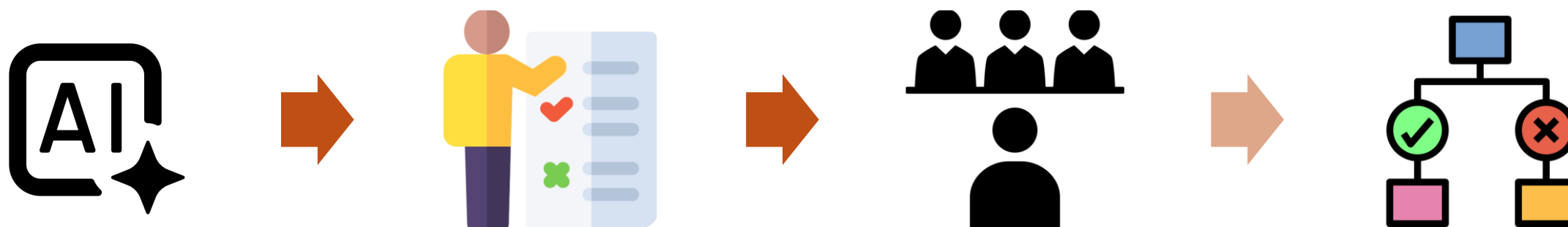
Small Sample:

Manual Content Analysis

Big Sample:

Automatic Content Analysis.

Small Sample: **Manual** Content Analysis



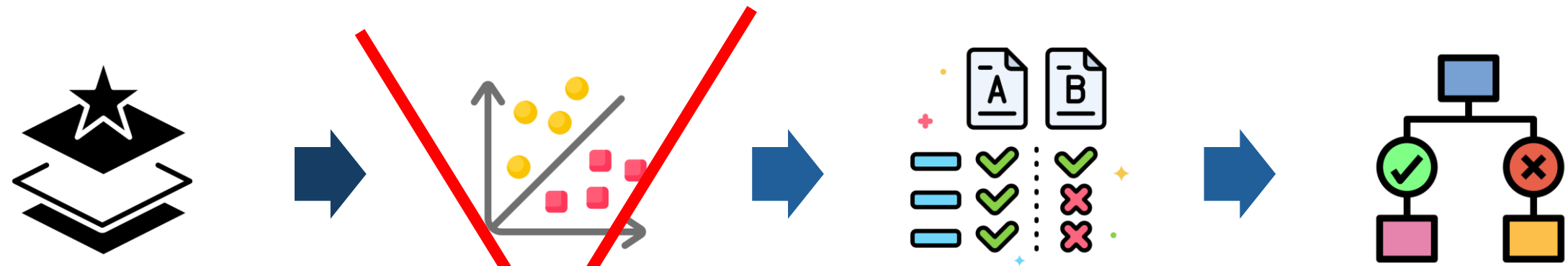
Produce a
measure for the
concept.

Write a codebook,
i.e., the
instructions.

Look for coders
and perform
intercoder
reliability.

Using the reliable
codebook finish
classifying the
data.

Big Sample: **Automatic** Content Analysis



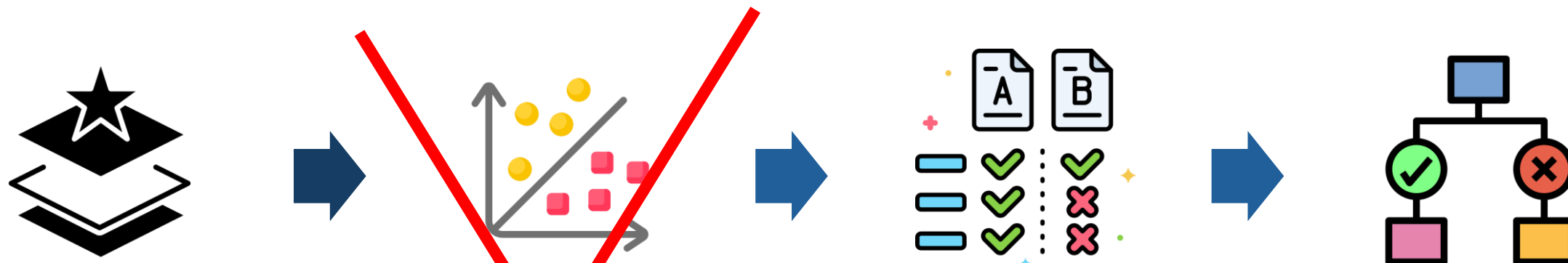
Using **Manual** Content Analysis produce a **Gold Standard**.

Use the Gold Standard to train a machine learning model or an AI agent.

Use Gold Standard to validate model or AI agent.

Using the best model/ agent classify all the data.

Big Sample: **Automatic** Content Analysis

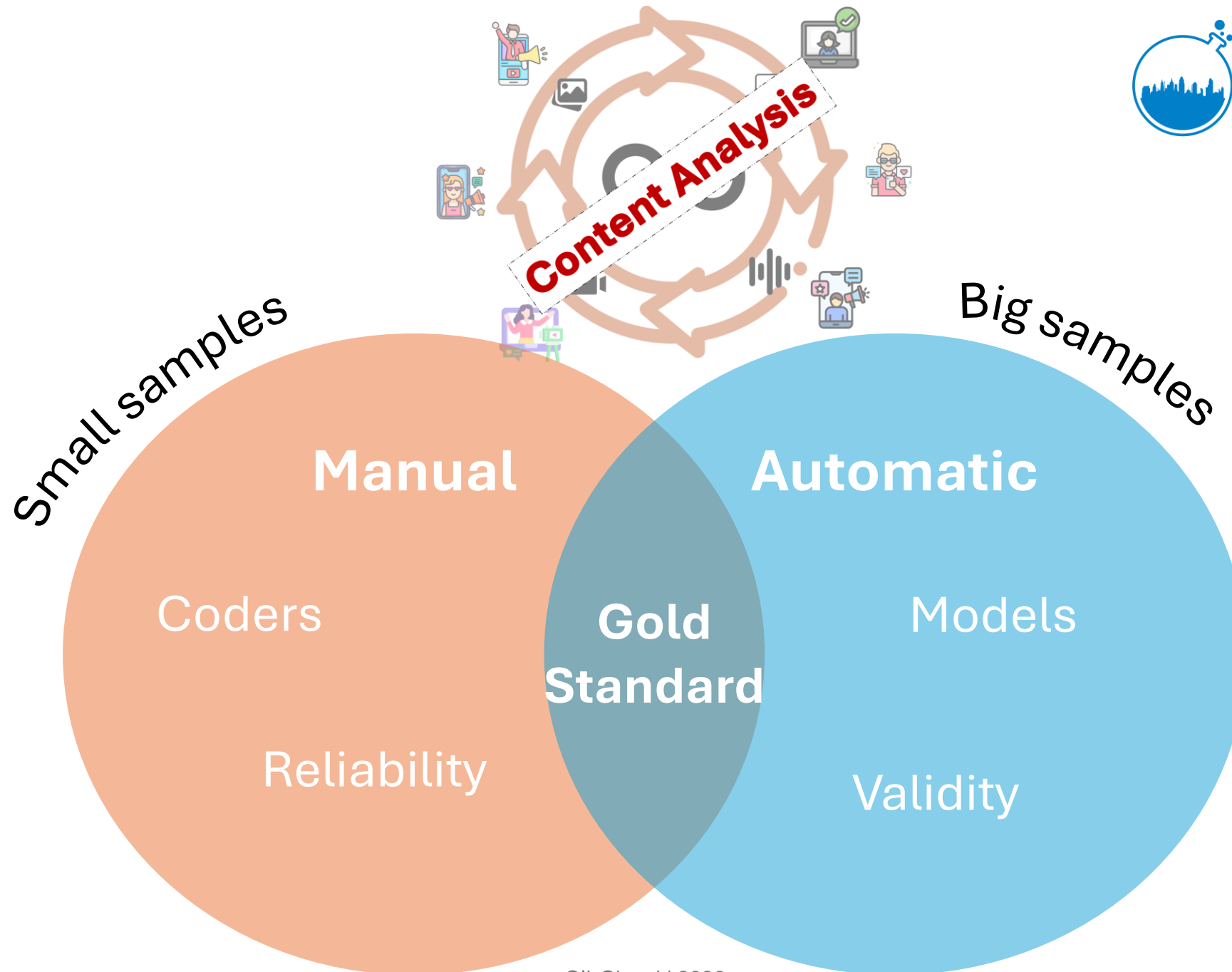


Using **Manual** Content Analysis produce a **Gold Standard**.

Use the Gold Standard to train a machine learning model or an AI agent.

Use Gold Standard to validate **zero-shot** AI agent.

Using the best model/ agent classify all the data.





Today, we will perform an automatic content analysis of the Guardian news about AI.

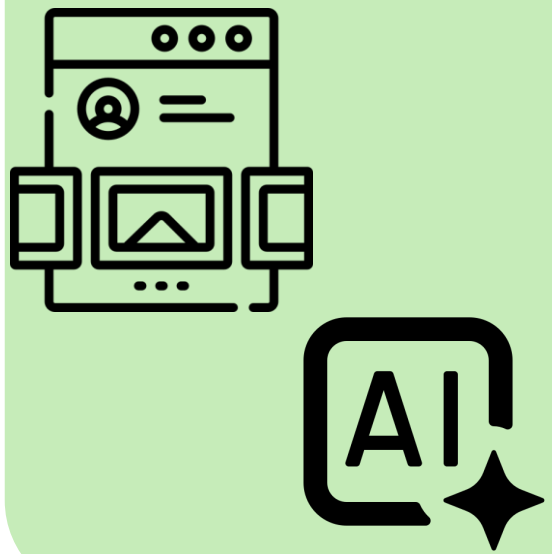
We will answer the question:

How has AI been portrayed in the Guardian news in the last 10 years?



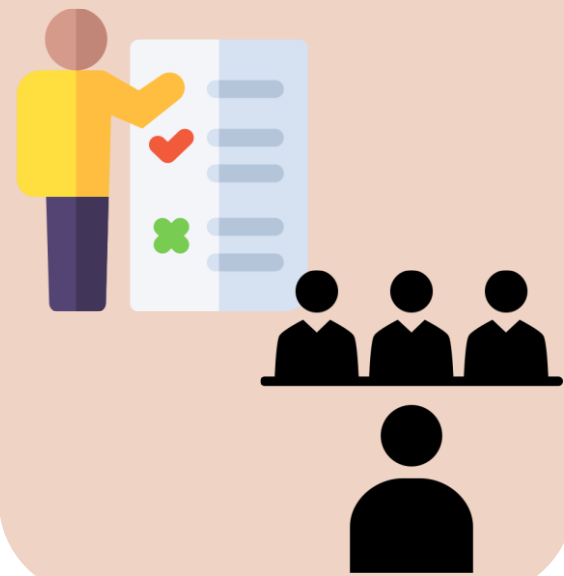
How has AI been portrayed in the Guardian news in the last 10 years?

1. Retrieve



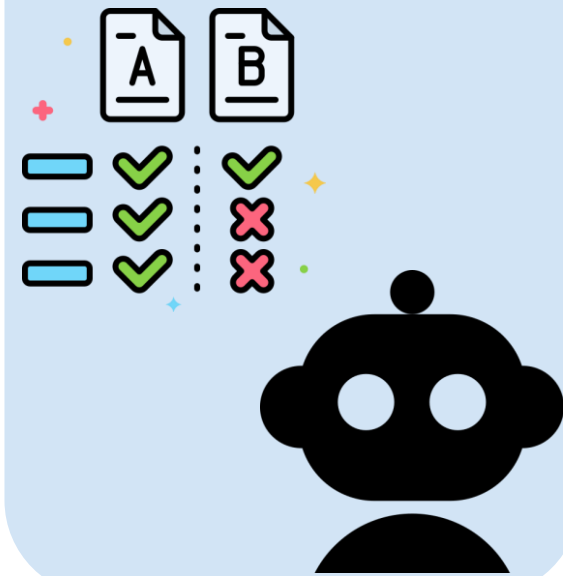
40 mins

2. Gold Standard



40 mins

3. Classify



40 mins

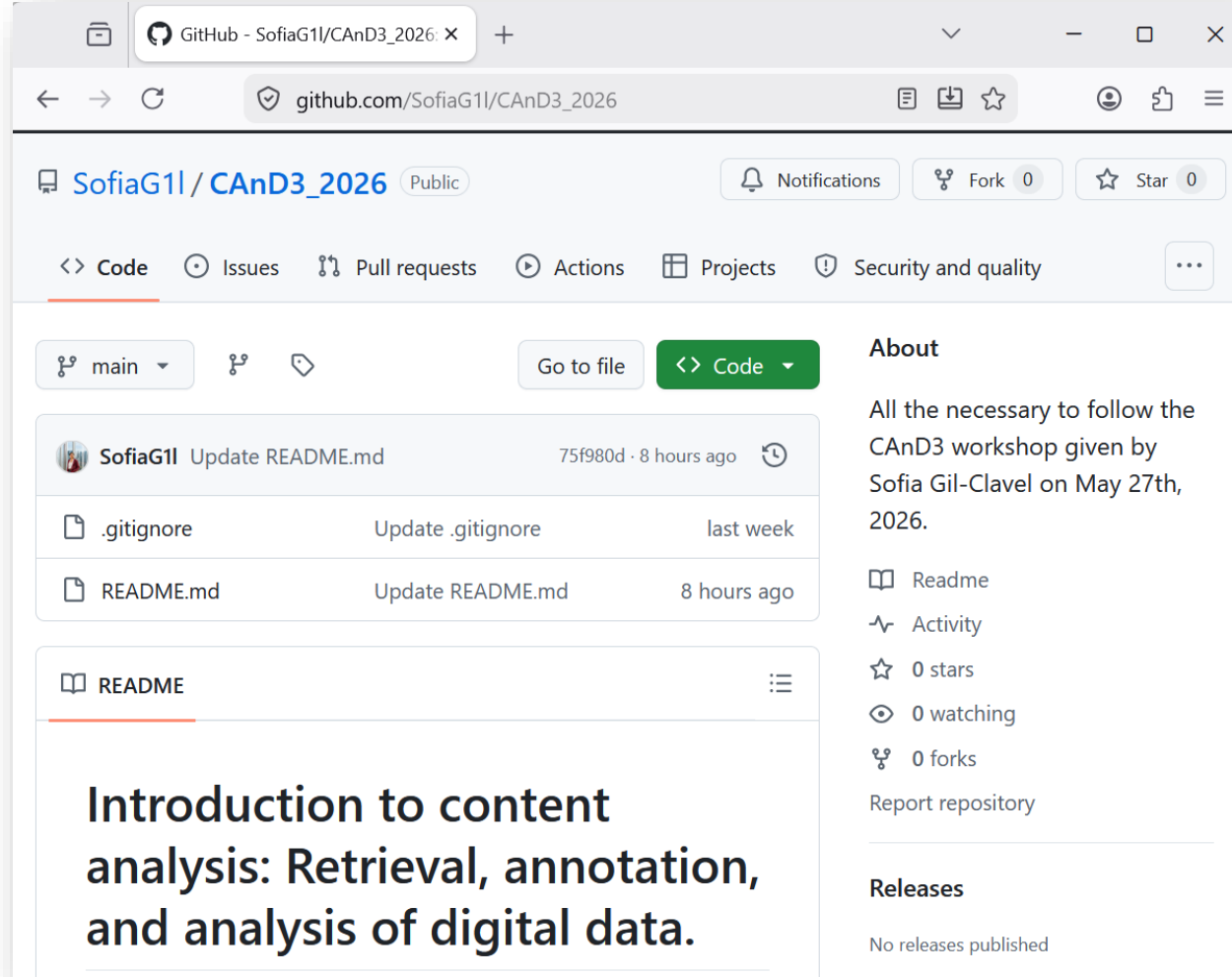
4. Analyze



40 mins

10 mins break



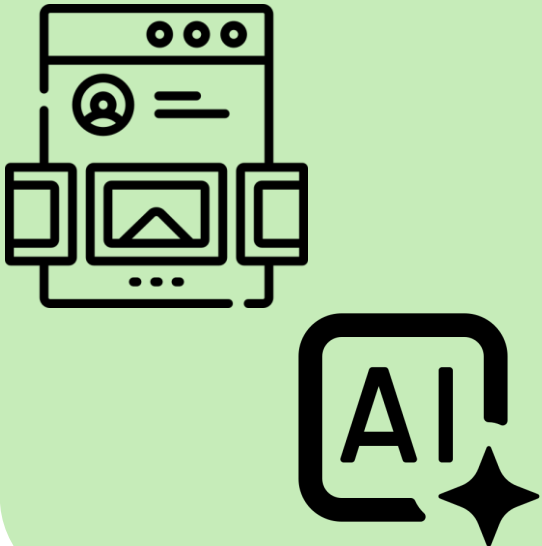


All the material is in
my GitHub:

https://github.com/SofiaG11/CAnD3_2026

How has AI been portrayed in the Guardian news in the last 10 years?

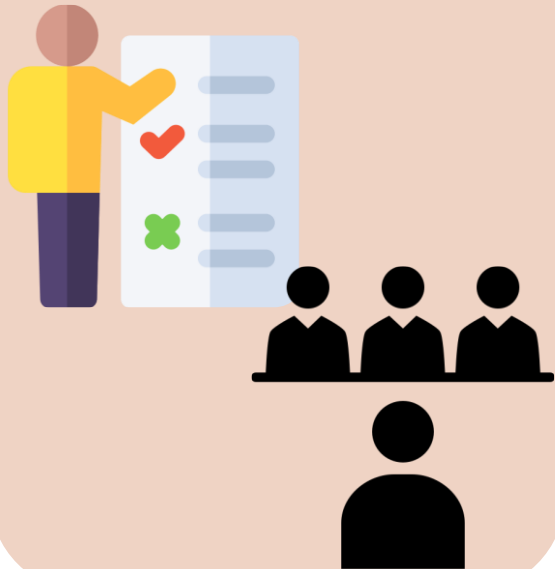
1. Retrieve



1. We will retrieve the AI Guardian news for the period 2014 to 2015.
2. Then, each participant will retrieve the data for a 6 months interval.
 - "from-date" = "2014-01-01", "to-date" = "2014-06-30"
 - "from-date" = "2014-07-01", "to-date" = "2014-12-31"
 - "from-date" = "2015-01-01", "to-date" = "2015-06-30"
 - "from-date" = "2015-07-01", "to-date" = "2015-12-31"
 - ...
3. You will send the RData to me via email:
B.S.GilClavel@vu.nl

How has AI been portrayed in the Guardian news in the last 10 years?

2. Gold Standard



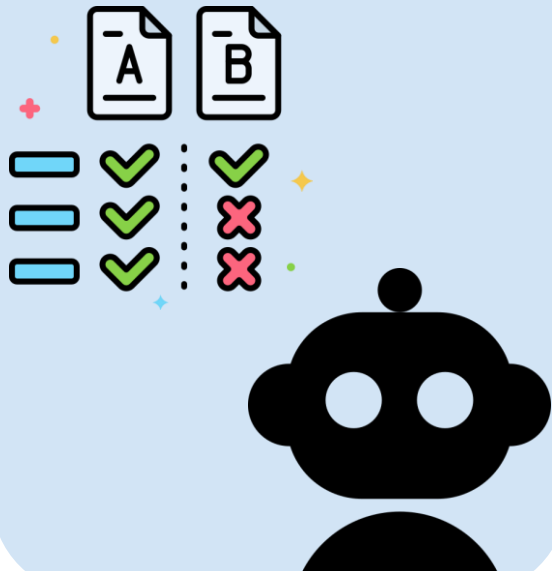
Codebook

Relevant	Irrelevant
Expresses concern related to AI risks.	Talks about new AI models released by the companies.
Expresses some type of feeling related to AI progress.	Describes AI usage.

1. Three people will be chosen as coders.
2. I will show them a sample of 10 news, and they will label them as relevant (**1**) or irrelevant (**0**).
3. We will calculate the intercoder reliability.
4. Each person will classify 10 more news base on the codebook.

How has AI been portrayed in the Guardian news in the last 10 years?

3. Classify



1. I will send you the full Rdata and the Gold Standard via email.
2. We will use the Gold validate the zero-shot classification of the AI agents.
3. We will classify the rest of the data using the best AI agent.

How has AI been portrayed in the Guardian news in the last 10 years?

4. Analyze



1. We will keep only the news classified as relevant.
2. We will perform some basic text analysis.
3. We will perform some basic text visualizations.